# SPRING BOARD VIRTUAL INTERSHIP 6.0

**NAME : SIVARAJ S**

**DOMAIN : DATA VISUALIZATION**

**PROJECT : VISUALIZING U.S NATURAL DISASTERDECLARATIONS**

**MENTOR : G.K.S JYOTEESH**

**Mile Stone 2 : Temporal Trend Analysis**

## 1.Objective

The objective of this project is to analyze and visualize historical disaster declaration data from the FEMA Disaster Declarations dataset to identify meaningful patterns and trends. The project focuses on understanding how disaster occurrences change over time, identifying the most frequent types of disasters, and detecting seasonal variations in disaster declarations.

Through data visualization techniques such as line charts, bar charts, and trend analysis, the project aims to provide clear insights into temporal patterns and disaster behavior. These insights help in understanding the frequency and distribution of disasters and demonstrate how data visualization can support better interpretation of large datasets.

## 2.Dataset Description

The dataset used in this project is the FEMA Disaster Declarations Summaries dataset, which contains records of federally declared disasters in the United States. It includes important information such as the date of declaration, type of disaster, and affected state.

Each record in the dataset represents a disaster event declared by FEMA. The dataset helps analyze disaster trends over time, frequency of different disaster types, and seasonal patterns through data visualization techniques.

## 3.Data Cleaning and Preprocessing

Before starting the analysis, several cleaning and preprocessing steps were performed on the dataset to ensure the data was accurate and suitable for visualization.

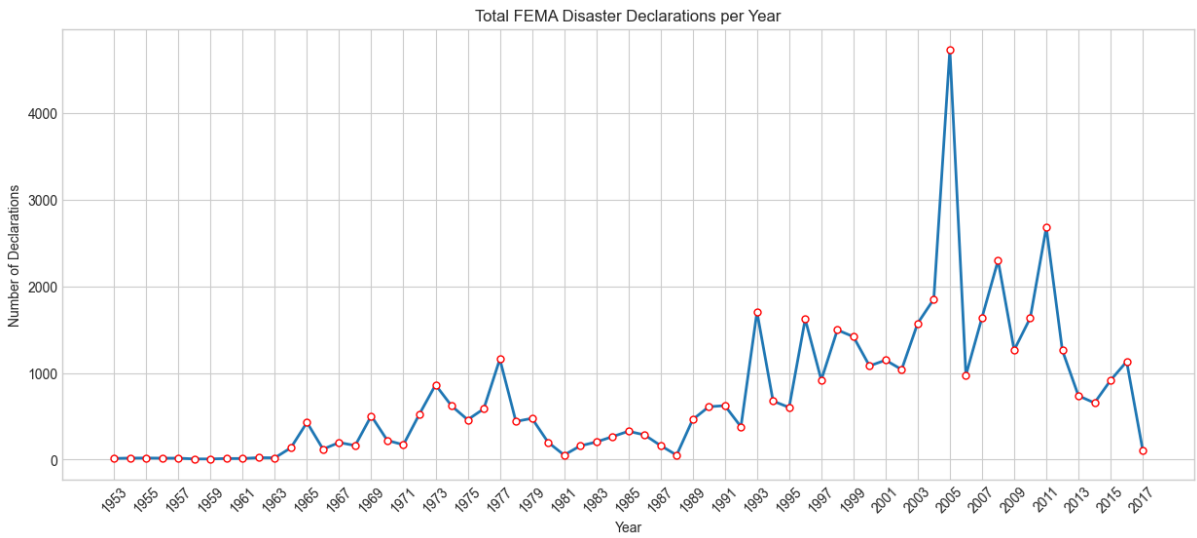
First, the declarationDate column was converted from string format to datetime format using data processing tools so that time-based analysis could be performed correctly. After converting the date column, two new columns — year and month — were extracted from the declarationDate field. This helped in analyzing yearly trends and monthly seasonal patterns.

Next, the dataset was checked for missing values and invalid entries. Any incomplete or incorrect records were handled to avoid errors during analysis. Duplicate rows were also identified and removed to ensure that each disaster declaration was counted only once.

Additionally, categorical values such as incidentType were reviewed and standardized to maintain consistency in naming across the dataset. These preprocessing steps ensured that the dataset was clean, structured, and ready for further analysis and visualization.

## 4.Yearly Trend Analysis

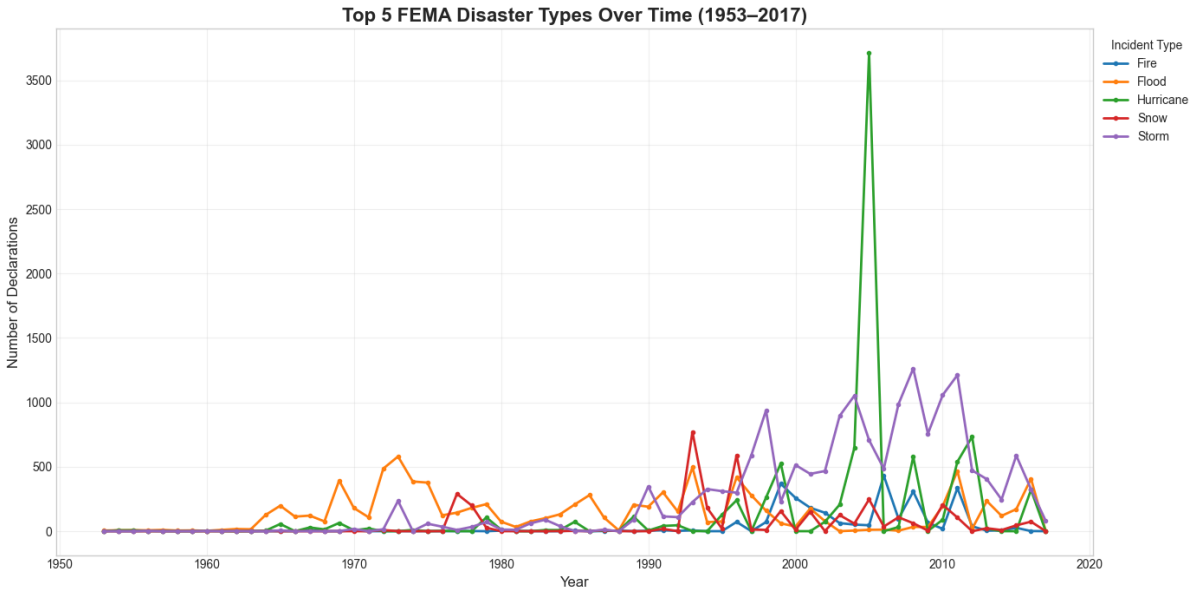
The dataset was grouped by year to calculate the total number of disaster declarations per year. A line chart was used to visualize the trend of disaster occurrences over time. This helps identify whether disasters are increasing or decreasing over the years.




- The graph shows the **total number of FEMA disaster declarations per year from 1953 to 2017**.
- There is an **overall increasing trend** in disaster declarations, with a major spike around **2005**.
- The increase may be due to **more extreme weather events, climate change, and improved disaster reporting systems**.

## 5.Incident Type Trends Overtime

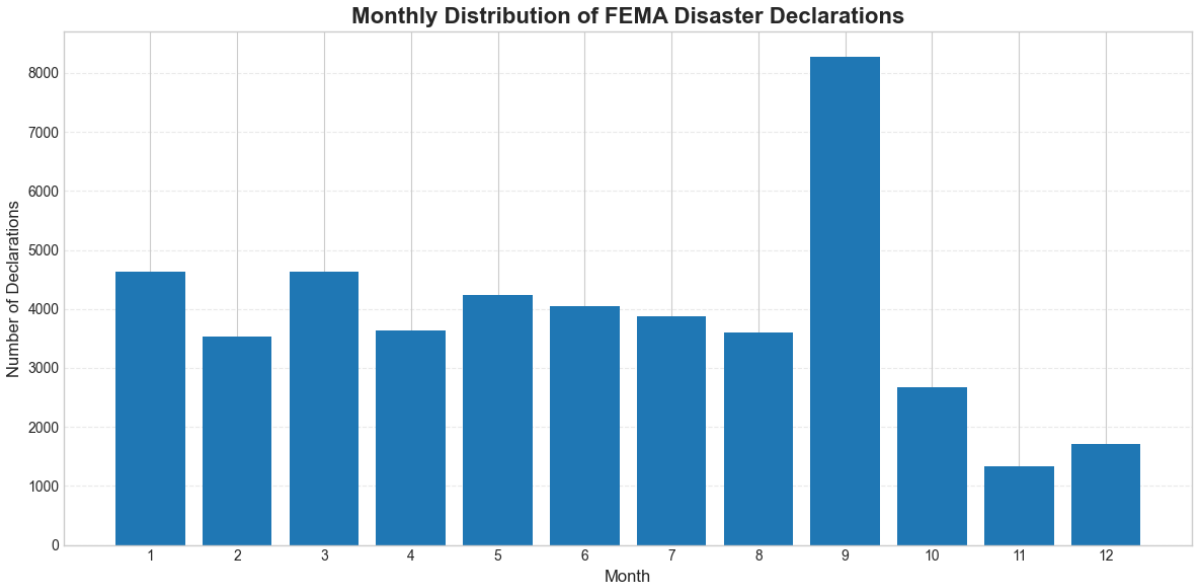
The dataset was grouped by year and incident type to analyze how different types of disasters changed over time. A multi-line chart was used to visualize the trends of various disaster categories across different years. This helps identify which disaster types occur more frequently and how their patterns evolve over time.




- The graph shows the trend of the top 5 FEMA disaster types (Fire, Flood, Hurricane, Snow, Storm) from 1953 to 2017.
- Storms and floods occur more frequently, and there is a large spike in hurricanes around 2005.
- This may be due to extreme weather events, climate patterns, and major hurricanes occurring during certain years.

## 6.SEASONALITY ANALYSIS (MONTHLY TRENDS)

The dataset was grouped by month to analyze how disaster declarations are distributed throughout the year. A bar chart was used to visualize the number of disasters occurring in each month. This helps identify seasonal patterns and months with higher disaster activity.

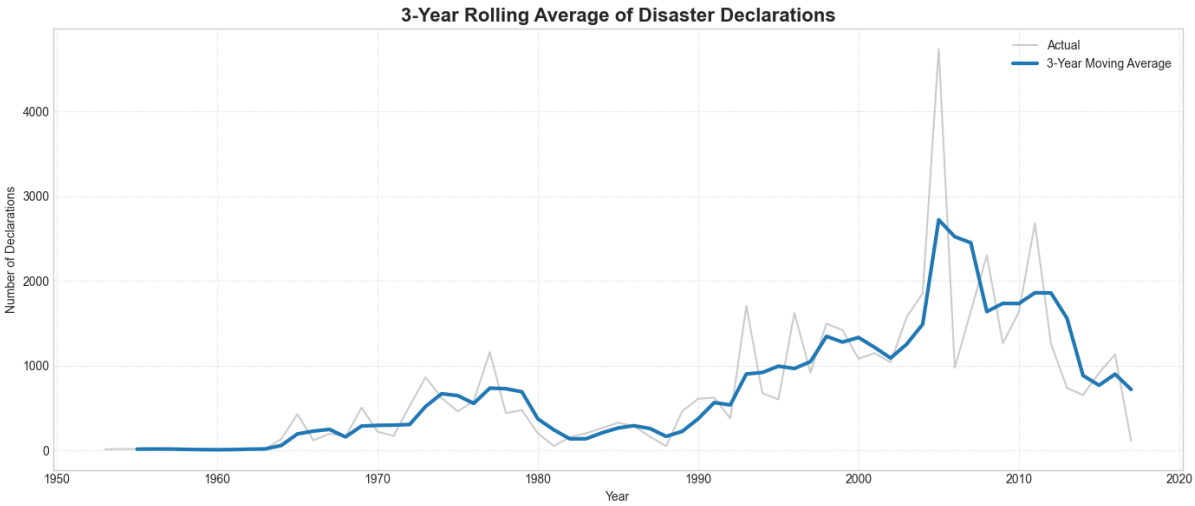



- The graph shows the number of FEMA disaster declarations for each month of the year.
- September has the highest number of disasters, while months like November and December have fewer declarations.
- This may be due to seasonal weather patterns such as hurricane season and extreme weather events occurring during certain months.

## 7. ADVANCED TIME INSIGHTS

### 7.1 ROLLING AVERAGE

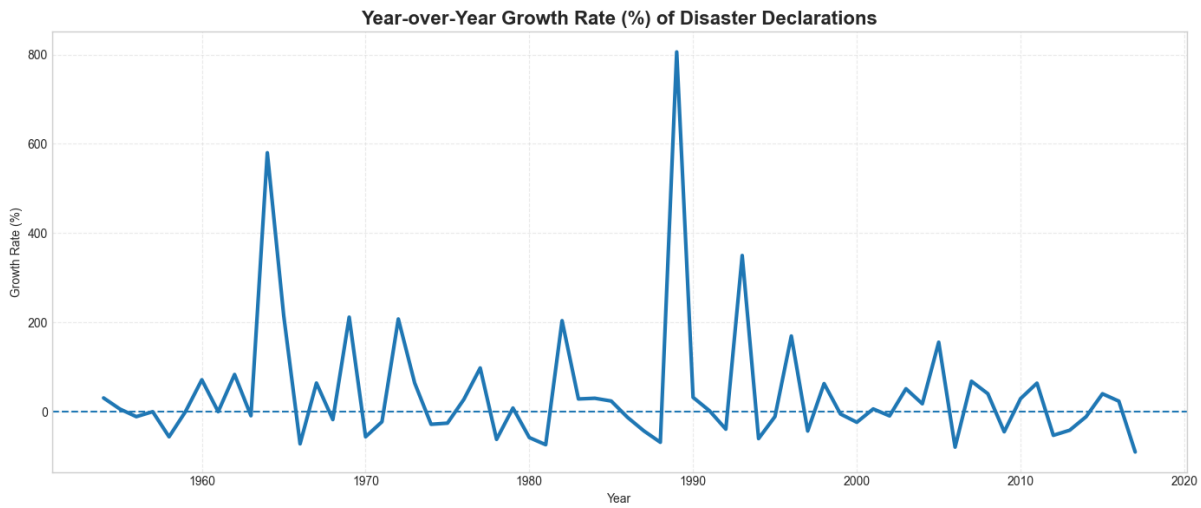
The dataset was analyzed using a 3-year rolling average to smooth short-term fluctuations in disaster declarations over time. A line chart was used to visualize the rolling average, which helps highlight the long-term trend in disaster occurrences more clearly.



- The graph shows the **3-year rolling average of FEMA disaster declarations over time**, compared with the actual yearly values.
- The rolling average shows a **clear upward trend over the years**, smoothing out the sharp fluctuations in the actual data.
- This trend may indicate an **increase in disaster occurrences over time due to climate change, environmental factors, and improved reporting systems**.

### 7.2 GROWTH RATE

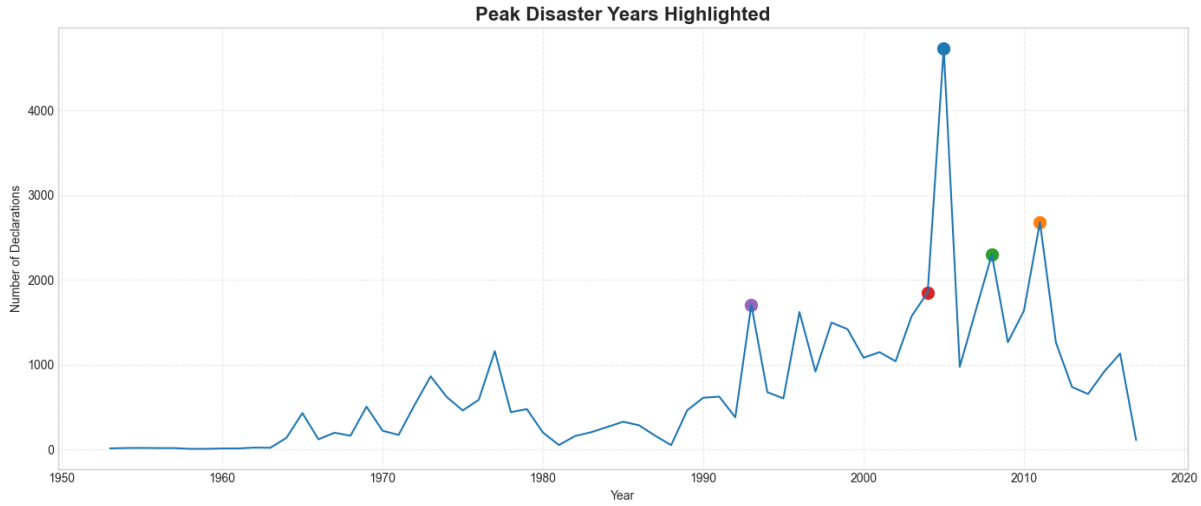
The year-over-year growth rate of disaster declarations was calculated to understand how the number of disasters changes from one year to the next. A line chart was used to visualize the percentage increase or decrease in disaster declarations over time. This helps identify years with significant increases or decreases in disaster activity.




- The graph shows the **year-over-year growth rate (%) of disaster declarations**.
- The growth rate **fluctuates significantly**, with some years showing sharp increases and others showing decreases.
- This variation may occur due to **major disaster events in certain years and fewer disasters in others**, causing sudden changes in yearly declarations.

### 7.3 PEAK DETECTION

The dataset was analyzed to identify the year with the highest number of disaster declarations. A visualization was used to highlight the peak year, which represents the period with the most disaster activity. This helps identify extreme years with unusually high disaster occurrences.




- The graph shows the **years with the highest number of FEMA disaster declarations**, highlighting peak disaster years.
- There are **clear spikes in certain years**, with **2005 showing the highest number of disaster declarations**.
- These peaks may occur due to **major natural disasters or multiple extreme weather events happening in the same year**.

### 8. CONCLUSION

The analysis of FEMA disaster declarations reveals a clear increasing trend in disaster occurrences over time. The visualizations show that certain disaster types such as storms, floods, and hurricanes occur more frequently, and disasters also display seasonal patterns during specific months. Additionally, some years show significant spikes in disaster declarations, indicating periods of extreme disaster activity.

Overall, the analysis highlights important temporal trends and patterns in disaster events, demonstrating how data visualization can help better understand historical disaster behaviour.